

Therapeutic use of curcumin-encapsulated and curcumin-primed exosomes

Journal of Cellular Physiology 2019

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Journal: Journal of Cellular Physiology

Year: 2019

Citation as given: J Cell Physiol. 2019;234(6):8182-8191

DOI: 10.1002/jcp.27615 | **Link:** <https://doi.org/10.1002/jcp.27615>

Abstract

Curcumin, the bioactive pigment of turmeric which has polyphenolic-hydrophobic components, has been used for the treatment of a variety of diseases. However, due to its insignificant intestinal-liver metabolism, low stability, quick systemic elimination and its hydrophobic property with low solubility, curcumin has limited bioavailability. Exosomes are nanovesicles (30–100 nm) released from diverse cell types into extracellular and, ultimately, into bio-fluids in a tightly regulated manner. Exosomes are capable of transferring lipids, proteins, RNAs and DNAs, both with and without direct cell-to-cell contact. Curcumin-encapsulated exosomes are highly bioavailable, soluble and safe, and can reach high concentrations in the blood; they, therefore, have therapeutic potential without toxic effects and immune stimulation. Thus, curcumin-encapsulated exosomes could be superior to other synthetic nanoparticles as a carrier of curcumin. The aim of the current review is to offer an overview of the in vitro, in vivo and clinical studies pertaining to the role of curcumin-primed and curcumin-encapsulated exosomes in the treatment of cancer, oxidative stress, brain disorders, cholesterol, and endothelial dysfunction.